



A MULTI-PLAYER GAME BY **amdon**

FACILITATION GUIDE





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1. Inquiry-based Learning: An Overview

Defining Inquiry

Inquiry in learners involves the posing of a researchable question and may be broadly categorized into 4 levels in varying ranges of sophistication demonstrated in learners.¹ For the purpose of this project, we have broadly listed in the table below the key features of activities that may promote the various levels of inquiry in learners:

Key features of each Inquiry Level and related activities			
Level 1	Level 2	Level 3	Level 4
Data provided, students guided on how to analyse and make connections between evidence available and observed phenomenon.	Data provided, students NOT told how to analyse Teacher-initiated questioning on the various elements of thought.	Lead-on from Level 2 Student-initiated (versus teacher-initiated) questioning one's own assumptions and testing them further.	Student-initiated questions and methods to explore them

Table 1: Key features of activities that may promote various levels of inquiry²

¹ National Research Council (NRC), 2000. *Inquiry and the national science education standards*. Washington, DC: National Academy Press

² Adapted from: Eick, Meadows & Balkcom, 2005, *Breaking into Inquiry: Scaffolding supports beginning efforts to implement inquiry in the classroom*. *The Science Teacher*: 72(7): 49-53



2. Key Features of Gut Feel™

The following features in Gut Feel™ enable the teacher to promote inquiry in learners during the game play:

- Presence of discrepant events.
- Availability of data for learners to analyse and draw conclusions on the various biological processes that take place in the alimentary canal.
- Competitive scenario-based games that challenge learners to apply concepts of human nutrition in order to outperform the other teams.
- Multi-user features that challenge learners to work in teams and collaborate with each other during the game play.

The tables below illustrate how the teacher may make use of discrepant events and data in the game to promote inquiry in learners. They also contain a list of possible questions / statements that the teacher can pose to learners during the review phases of the game play to promote understanding of specific processes in human nutrition.

2.1. Using discrepant events to promote inquiry in learners

Examples of possible discrepant events	Suggested teaching strategies	Questions / Statements to probe for understanding
Learners, especially those in the Brain, Mouth, Oesophagus and Stomach, see that George has been eating a lot of food but the nutritional level in his body doesn't seem to increase significantly.	Allow the game to progress, constantly monitoring George's nutritional level in each team and commenting on whether a particular team is achieving its target nutritional level or if the team's diet is skewed towards specific food types.	▪ What was in your mind when you chose the type of food to eat? ▪ What were you expecting to automatically happen (or what did you assume to happen) once you have eaten and swallowed the food? ▪ How can we ensure that the food we eat actually contributes to the total nutritional level in George's body?



Examples of possible discrepant events	Suggested teaching strategies	Questions / Statements to probe for understanding
Learners expecting to swallow down as much food as possible to do well will instead cause problems for themselves due to implementation of food quotas at the oesophagus area. Messages will inform the learners when quota has been exceeded. Exceeding the quota will lead to penalties in the score.	Allow learner to continue game play; intervene only when the quota has been exceeded and penalties have started. Help learners realise that the oesophagus is not a hollow tube but one that is controlled by muscles that contract rhythmically to push the food down towards the stomach.	- Some of you are experiencing difficulties in swallowing the food that has been chewed. Think about where the food is to go next and check if the food is stuck somewhere in the throat. - Do you think that by just eating a lot of food and swallowing it will automatically resulting in an increase in nutritional level in George's body? What needs to happen to convert the swallowed food to nutrients in the body? What do you need to do next to ensure that this happens?
Learners observe that the food cannot be swallowed unless some saliva has been applied to it first.	Allow the learners to discover that saliva helps to lubricate the food, thereby aiding in swallowing. Ask learners to identify and guess if the food needs to be lubricated to facilitate swallowing.	- Why do you think the food cannot be swallowed even after it has been broken down? - How can you make the food more moist without having to drink anything? - Why do you think it is important the food is moist before swallowing? Can you predict what will happen in the oesophagus if the food is swallowed without first being lubricated?
Learners observe that performing poorly in rhythmic activities at the oesophagus gives rise to food motility issues while points are awarded for when a perfect rhythm is implemented.	Allow learners to discover that implementing a poor rhythm on the peristalsis activity will slow down the speed of the movement of food bolus down the oesophagus.	- How can you increase the speed of food movement down the oesophagus? - Is there a rhythm or pattern to which the different regions in the oesophagus may contract to push the food down? - Do you think this process happens naturally in our oesophagus (ie. We do not need to actively contract the muscles in the oesophagus to push the food towards the stomach)?



Examples of possible discrepant events	Suggested teaching strategies	Questions / Statements to probe for understanding
Learners observe that implementing a poor rhythm on the peristalsis activity at the churning control in the stomach will slow down the mechanical breaking up of the food grid into smaller sections. Abstract 'food grids' are then used to represent food composition & to demonstrate the importance of surface area for digestion in the stomach. Learners have to implement more churns to mechanically break up the food grid if the food is swallowed hastily and is not reduced in hardness from the mouth.	Allow learners to discover that implementing a poor rhythm on the peristaltic churning activity will slow down speed for breaking up the food mechanically. Allow learners to discover that the stomach has to work more and harder to break up the food if the food is high in hardness level.	- Did you notice which parts of the food particle are typically broken up by the enzymes? Why do you think the enzymes tend to attack only these parts of the food particle? - How can you increase the rate of food break down by enzymes in the stomach? - How does churning in the stomach help increase the rate of breakdown? - Does it mean that its easier to digest semi solid foods than solid foods, assuming that they both contain the same type of nutrients?
Learners (in the Stomach Room) discover that food particles do not break up as they may have expected unless Hydrochloric acid (HCl) is released into the stomach.	Allow learners to progressively notice that food particles do not seem to break up significantly initially despite the appropriate churning motions and enzymes release. Extension: tell learners about the corrosiveness of hydrochloric acid and how it is actually very acidic in the stomach; then challenge them to infer the impact of having too much acid in the stomach.	- Look around the Stomach Room, are there other controllers that need to be managed? - Is there something missing such that the enzymes cannot do their functions? - Can you guess a possible function of Hydrochloric acid?
Learners need to form a continuous pipeline to release HCl into the stomach. Learners observe that the pepsinogen enzymes released in the stomach do not break down the proteins when the pH level is not reduced to 2.	Allow learners to progressively notice that enzymes released do not seem to react with any of the exposed proteins despite the appropriate churning motions and enzymes release.	- Let's imagine for a moment that George stops eating but there's still a lot of Hydrochloric acid in the stomach. Now Hydrochloric acid is corrosive (as mentioned earlier), what do you think can happen to George's stomach if this situation continues? - Look around the Stomach Room, are there other controllers that need to be managed? - Is there something missing such that the enzymes cannot do their functions? - Can you guess a possible function of Hydrochloric acid?



Examples of possible discrepant events	Suggested teaching strategies	Questions / Statements to probe for understanding
Learners are prevented from lowering stomach acidity below pH 2 by the locking of the hotseat.	Extension: tell learners about the corrosiveness of hydrochloric acid and how it is actually very acidic in the stomach; then challenge them to infer the impact of having too much acid in the stomach.	Let's imagine for a moment that George stops eating but there's still a lot of Hydrochloric acid in the stomach. Now Hydrochloric acid is corrosive (as mentioned earlier), what do you think can happen to George's stomach if this situation continues?
Learners observe that activated pepsin enzyme reacts only with the exposed proteins of the food grid.	Allow learners to know that mechanically breaking up the food into smaller pieces increases the surface area exposed to the enzymes. This will allow the enzymes to react with the proteins effectively.	- Did you notice which parts of the food particle are typically broken up by the enzymes? Why do you think the enzymes tend to attack only these parts of the food particle? - How can you increase the rate of food break down by enzymes in the stomach? - How does (a method suggested by learners, such as mechanically breaking up food into smaller pieces) help increase the rate of breakdown? - Does it mean that its easier to digest semi solid foods than solid foods, assuming that they both contain the same type of nutrients?
Learners observe that in some organs in the alimentary canal, the enzymes released do not react with the fats and carbohydrates in the food.	Allow the learners to discover that only proteins are digested in these organs.	- Are all kinds of nutrients (carbohydrates, proteins and fats) broken down in the stomach? - Do you think there is a possible advantage of digesting only proteins in this organ?
Learners attempting to use turret 1 without first gathering the enzymes from the pancreas and bile from the gall bladder will be unable to digest the incoming food.	Remind learners to collect the enzymes and bile from the correct organs, allow learners to explore and discover that the enzymes are produced from the pancreas and bile is stored in the gall bladder.	- Pay attention to the type of enzymes or substances that you collect from the pancreas and the gall bladder. - See if you can identify what type of substances will react with which type of food in the gut.
Learners observe that protein substrates have to be broken down twice with 2 different types of enzymes, firstly at the stomach or at turret 1 at the small intestine and secondly at turret 2.	Where necessary, alert learners to the possibility of two-staged break down of proteins and carbohydrates. Ask learners to guess any possible reasons for a two-staged process.	- Where are proteins and carbohydrates digested in the alimentary canal? - Is the food that is digested only in the stomach or small intestine absorbed? Why do you think this is so? - What do you think will happen to the



Examples of possible discrepant events	Suggested teaching strategies	Questions / Statements to probe for understanding
Learners observe that carbohydrate have to be broken down twice with 2 different types of enzyme, firstly at turret 1 at the small intestine and secondly at turret 2.	For higher ability learners, you may wish to challenge them to remember the type of enzymes needed to break down proteins and carbohydrates in each stage and the product of each stage of digestion.	protein and carbohydrate present in the food if they are only digested by enzymes in turret 1 (or 2)? Will the products of the digestion be absorbed? - How do you ensure that the food is actually digested to the point that it can be absorbed? - Do you remember the type of enzymes that digest the carbohydrates and proteins? - Do you think there is a difference in physical conditions in the stomach and the small intestine (eg. pH difference)? - Considering that enzymes work well only at suitable pH, is it better (or worse) to have all the digestion take place in just one organ (eg. Stomach) rather than in the stomach and the small intestine?
Learners observe that fats can be digested with lipase alone at turret 1 at the small intestine. However fats that have reacted with bile first before being broken down by lipase generates a higher score.	Learners realise that fat can be digested without using bile when they mistakenly hit it with lipase. Explain that emulsification is intended to increase the efficiency of fat digestion but does not break down fat chemically. The notions of the benefits of the emulsification of fats before digestion is suggested and reinforced with the additional scores awarded.	- How many times must fats be digested before it is ready for absorption? - Did you notice that some chemicals, with mixed with the fats can increase the rate of digestion? Where are these chemicals found? - What do you think is the function of the gall bladder? - Will the person die if his gall bladder is removed? - If you are a dietician, what would you recommend to the patient who lacks a gall bladder? - Why is eating less fatty foods beneficial to a patient who lacks a gall bladder?
Learners observe that scores are penalised for shooting the enzymes at the wrong substrates or when missing and wasting enzymes by shooting aimlessly.	Remind learners that enzymes are highly specific and will only react with their specific substrates.	- Do the enzymes react with all the substrates as they travel down the gut? - Did you notice that your scores are penalised when you shoot the enzymes at the wrong substrates? What does



Examples of possible discrepant events	Suggested teaching strategies	Questions / Statements to probe for understanding
Learner observed that nutrients not broken down at turret 1 (by the pancreatic enzymes) couldn't be broken down even when the correct enzymes are used at turret 2.	Remind learners to cooperate and distribute tasks with teammates to operate the digestion controls upstream to ensure that the nutrients that arrive at the turret 2 control been broken down appropriately.	this say about the enzymes with regard to the substrates they react with? - Can you the types of enzymes and the name of the substrates they react with? - What do you think can happen to us with we lack one type of enzymes? If you are a dietician, what would you recommend to the patient?
Learners need to punch the fully broken down nutrients into the pipe-like channels for absorption at the absorption controls. Learners observe that nutrients that are not fully broken down to their simplest form (in this game, the simplest forms assume the shape of a diamond) are not able to pass into the absorption channels and any attempt to do so will yield negative scores and will not contribute to the nutrition-o-meter.	Remind learners to cooperate and distribute tasks with team mates to operate the digestion controls upstream to ensure that the nutrients that arrive at the absorption control have been fully broken down. Ask learners to think about why nutrients that are not fully broken down cannot enter through the channels	- Did you discover that in some instances, your attempts to punch the nutrients into the absorption channels resulted in negative scores? What do you think might have happened? - Can partially-digested food be absorbed? - Do you think the person in this room plays an important role? What happens if no one is manning this room? - What do you or your team needs to do to ensure maximum efficiency in food absorption? - How do you think nutrients are absorbed in nature?
Learners observe that fibre is not broken down nor absorbed anywhere along the gut.	During game play, ask learners to pay attention to the fate of fibres in the gut (learners may need to reference their peers in the brain on the food that is ingested in order to identify those that are high in fibre). Assign bonus points to learners who can correctly suggest the fate of fibre in the gut and the role of fibre in human nutrition.	- Is all the food that is eaten digestible by enzymes in the human body? Can you name one type of food that is eaten is not digested? - What then do you think is the function of fibre in our diet?



2.2. Using data to promote inquiry in learners

Data in Gut Feel™ is presented in the following ways:

- As points awarded (or deducted) when certain actions are performed appropriately (or inappropriately).
- As data tables that display key information for players to analyse and make decisions
- As information bars that fluctuate in real time as the game progresses, prompting learners to take the necessary corrective actions
- Indirectly in the form of smooth transition from one compartment of the alimentary canal to another. Conversely, a disrupted flow of food from one compartment to the next constitutes a discrepant event, which is discussed in the previous section.

The following table illustrates how the teacher can make use of data presented on screen to engage learners and facilitate understanding during game play.

Illustrative example	Suggested teaching strategies	Questions / statements to probe for understanding
Learners observe that a small amount of carbohydrates is digested by enzyme in the saliva applied. Points are awarded.	Draw learners' attention to the points awarded when carbohydrates is digested in the mouth as learners are not likely to notice this.	▪ Did you notice that points are awarded when you release saliva? ▪ What do you think these points are awarded for? ▪ Did you notice that some of the food is digested when saliva is released? ▪ What to you think is the function of saliva in this process?
Learners observe that implementing bite points nearer to the molars of the teeth reduces the hardness level of the food more compared to points nearer to the canine or incisors of the teeth.	Ask learners to observe the efficiency of each bite when each bite is initiated at different zones in human dentition. Encourage learners to maximise each bite and aim for the regions in the teeth that can bring about maximum efficiency to break down food.	▪ Every bite counts; how to ensure the highest bite points when a bite is initiated? ▪ Why does implementing bite points nearer to the molars reduce the hardness of the food the most? ▪ Do you think molars are adapted for chewing? How?
Learners observe that the lower the hardness level of the food at the point that the food is swallowed, the higher the score awarded.	Alert learners to the score that is awarded in the mouth room, with respect to the hardness of the food. Challenge learners to guess the reasons for higher scored with decreased hardness of the food.	▪ Why do you think swallowing a softer food can earn higher scores? ▪ How do we ensure that the food is soft enough to swallow? Is biting alone enough?

Illustrative example	Suggested teaching strategies	Questions / statements to probe for understanding
Colour-coded enzyme particles help learners associate the type of enzymes to the substrate (exact names of enzymes are only expected at secondary levels). Changes in visual representation of substrates shows the effect of digestion by pancreatic enzymes (turret 1) and intestinal enzymes (turret 2)	Remind learners in the pancreas, stomach and small intestine to take note of the colour of enzyme bullets that are being fired off (in the stomach and small intestine). Encourage learners to constantly check with their teammates in the brain and mouth on the type of food that is being ingested in order to ensure adequate stock-up of the specific enzymes for digestion.	- Did you notice that enzymes are very specific in the substrates / food particles they target? - Remember to be very specific when aiming to shoot the food particles with different enzymes. - Remember not to focus on just firing off the enzymes at the food particles. You need to be specific. Look carefully before you shoot the enzyme to prevent negative points.
Nutritional tables: Choosing a low fat, high carbohydrate diet may still yield a fat, unhealthy George at the end of the scenario. Choosing a well balanced diet but poor digestion and absorption game play will still result in an under weight George at the end of the scenario.	Allow learners to conclude that excess carbohydrates absorbed will be converted to fats. Allow learners to discover the importance of proper digestion and absorption of food for the body to be able to utilize its nutrients for energy.	- What do you will happen to the excess carbohydrates that have been absorbed? Did you see any changes to George's body shape after a round of carbohydrate-heavy diet? - What kind of food is very rich in fat / proteins / carbohydrates? - Looking at George's example, what kind of food should you take daily in order to have a balanced diet? Why?
Wastage Bar: Wastage of food by not absorbing certain nutrients deliberately will result in subtraction of scores as penalties. Wastage meter bar keeps track of amount of nutrients wasted in the process.	Learners realise that any attempts to meet diet requirements through selective absorption of selected nutrients will lead to poorer scores. Remind learners that planning a suitable diet and digesting properly are the appropriate way to minimise wastage.	



Illustrative example	Suggested teaching strategies	Questions / statements to probe for understanding
Nutrition-o-meter: The Nutrition-o-meter in the game keeps track of nutrients digested and goals needed to attain the daily nutrition requirement. With three meals worth of play to meet the goals, student can adjust accordingly and choose foods that meet the requirement without resorting to wastage. The meter is also used to compute the scenario projection. Information on the Nutrition Table pertinent both to the game and also real life. Layout of the Table also closely follows that of actual labels on food products.	Remind students to constantly observe the Nutrition-o-meter to choose the appropriate food to meet their requirements.	- What do the bars on the Nutrition-o-meter tell you about the nutrient content in George's body? - Let's take an example: if the meter shows that the level of fat in George's body is way above the target level, what does this mean? What does it say about the diet you have chosen for George so far? What do you need to do next? If this happens in real life, what does the person need to do?
Feedback system: Feedback and messages contain learning points and inform the team of key developments in the digestive process.	Alert learners to the feedback messages and the scores throughout the game play as hints.	- Pay attention to the feedback and hints that the game releases to you from time to time. They contain very important information that can help you do well. - You will quickly realise that in order to do better than your peers, you need to think through (eg. What kind of food to eat, where to go next etc) and strategise. It is not about how fast you work but how well you apply your skills and knowledge of human digestion that counts.
Score system: Score system serve as rough guide as to whether students are doing right or wrong things. Penalties for performing actions that don't happen naturally in normal digestive system.		



3. Suggested Method of Use

We suggest using Gut Feel™ over a few lessons / phases, with each phase possibly fitting into a one / two-period lesson, depending on the time allocated per period. The table below briefly describes the activities and the suggested teaching strategies in each phase.

Phase One	
Activity	Learners introduced to the main controls in Gut Feel™ and explore manipulating these controls in Gut Feel™ Tutorial. Learners note the key activities that take place in each room (region of the alimentary canal).
Suggested Teaching Strategy	Ask the learners to list the organs that are responsible for the human body to obtain the necessary nutrients from food (ie. Organs that are involved in the ingestion, digestion and absorption of food in the human body). Ask learners to suggest the processes that take place in each of these organs. Introduce Gut Feel™ as a game; tell the story of George and the role of Automatons. Introduce the learners' role in Gut Feel™. Allow learners to explore Gut Feel™ Tutorial. Emphasize the need to observe carefully the activities that take place in each compartment. Mention that the game is best played when 4 or 5 players form one team, each managing different rooms at different times. After the learners have completed the tutorial: ▪ Ask learners to recall what they have observed and describe the activities that take place in each compartment in the alimentary canal. ▪ Ask learners to suggest ways to improve efficiency in each room and how the efficiency in each room is dependent on the decisions made by other players in the preceding rooms. ▪ Ask learners to form teams of 4 or 5 players for the next game. Announce that in the following lesson, each team will work together in Gut Feel™ to compete with the other teams in the class. Inform learners that you may release a series of scenarios or crises for learners to overcome, with each scenario getting more complex than the preceding one. Inform learners in advance that no learner is allowed to stay in the same room for more than 3 minutes (if discovered, the team will be disqualified) Let learners know that in order to do well in each round, each member of the team will need to: ▪ Be familiar with the function(s) of each organ in the human digestive system. ▪ Work well with the other members of the team. Refer learners to the textbook or the Gut Feel™ User Guide for some reference notes on the function of each organ and the key processes that take place in human digestion.



Phase Two: Life as it is	
Activity	Learners work in teams to compete with each other to help George attain the adequate nutrient target stipulated in the base scenario. Teacher reviews with learners the strategies to help learners do well in this game and relate it to nutrition in human beings.
Suggested Teaching Strategy	Emphasize the rule that no learner is allowed to stay in the same room for more than 3 minutes (if discovered, the team will be disqualified) Let learners work in teams and launch Gut Feel™ according to the instructions in the user guide. Allow 20 minutes of game play. During the review part of the lesson: - Ask learners if they have noticed any discrepant events: - Did they expect certain processes to take place but were surprised that these processes did not take place? - Were they surprised that some processes took place? - Did they discover the reasons for these events taking place during the game play? Can they relate these events (in the game) to what happens in our digestive system? - Did they do anything the future to prevent them from happening? - Discuss with learners specific discrepant events that might have occurred during the game play, using the table in the previous section ('Using Discrepant events to promote inquiry in learners') as reference. - Discuss with learners the relevance of data presented throughout the game, using the table in the previous section ('Using Data to promote inquiry in learners') as reference. - Ask learners to suggest strategies to be more effective in the game play. - Discuss with learners if they have effectively collaborated with each other and how they felt when their team members failed to cooperate. If time allows, let learners play one more round of the game, or you may wish to launch any particular scenario in the game to increase the level of difficulty. Should you decide to launch a scenario, refer to Phase 3 of in the following page for suggested teaching strategies.



Phase Three: Life in crisis (Extension)

Activity	Learners work in teams to compete with each other to help George overcome the crisis presented in each scenario. Teacher reviews with learners and help learners relate their effective strategies during game play to the concepts surrounding human nutrition in each scenario.
Suggested Teaching Strategy	Review with learners the parts of the human digestive system and the functions of each organ. Inform learners of the scenario that you wish to launch. Ask learners to discuss with their team mates: ▪ What does the scenario mean to George's digestive system? What is the ideal outcome at the end of this round in the game? ▪ What does George need to do to overcome the crisis? ▪ What does that mean for players in each room? ▪ How can the team do better than the other teams in the game? Conduct a post-game review, following a similar line of questioning as in Phase Two. Ask learners if the predictions they made before the game were correct, giving reasons. Extension: set learners an assignment by challenging learners to think through an additional scenario (set by the teacher and not found in Gut Feel™). Let learners work either individually or in teams to create a powerpoint presentation for discussion with the rest of the class in the following lesson.



4. About the developer
